The Turning Effect — Paper B

MEDIUM · 40 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Always convert centimetres to metres before you multiply, and give every moment the unit N m.
- The mark for each question is shown in square brackets on the right.
- Reminder: moment = force × perpendicular distance from the pivot.

SECTION 1 — BALANCING

1. State the principle of moments. [2]

2. Ravi, whose weight is 320 N, sits 1.6 m from the pivot of a see-saw. Meera weighs 400 N. How far from the pivot must she sit to balance it? [3]

3. A metre rule is pivoted at its 50 cm mark. A weight of 2 N hangs at the 20 cm mark.

(a) Calculate the anticlockwise moment about the pivot. [2]

(b) A 4 N weight is used to balance the rule. At which mark must it hang? [2]

4. A crowbar is used to lift a heavy stone. An effort of 200 N is applied 1.2 m from the pivot, and the stone sits 0.2 m from the pivot on the other side. Calculate the largest weight of stone that can be lifted. [3]

SECTION 2 — MORE THAN TWO FORCES

5. On the left of a see-saw pivot, a 250 N child sits 1.2 m away and a 150 N child sits 2.0 m away. A single child of weight 400 N sits on the right.

(a) Calculate the total anticlockwise moment. [2]

(b) How far from the pivot must the 400 N child sit to balance the see-saw? [2]

6. Explain each of these using the idea of moments.

(a) A door is harder to push open near its hinges than at its handle. [2]

(b) A loaded wheelbarrow is easier to lift when the load is placed close to the wheel. [2]

7. A see-saw is balanced. One child then slides closer to the pivot. State and explain what happens next. [3]

8. Calculate the moment of a force of 18 N acting 45 cm from a pivot. [2]

SECTION 3 — USING THE IDEA

9. Give two situations in everyday life where a large moment is useful, and explain one of them. [3]

10. A student writes: “The moment of the force is 30 N.” Explain what is wrong with this statement. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.